



PANDASIA
PREVENTING FUTURE PANDEMICS



D9.2 Protocol for VHV training assessment



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PANDASIA

Protocol for VHV training assessment

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Project Partners

No.	Abbreviation	Full name	Country	Main roles
1	CU	Chulalongkorn University	Thailand	WP1 lead
2	MU	Mahidol University	Thailand	WP2 lead
3	KKU	Khon Kaen University	Thailand	WP3 lead
4	UMU	Umeå University	Sweden	WP4 lead
5	UKHD	Universitätsklinikum Heidelberg University	Germany	WP5 lead
6	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
7	NMBU	Norwegian University of Life Sciences	Norway	The Coordinator (WP7)
8	QMUL	Queen Mary University of London	UK	WP1-Partner
9	NVI	Norwegian Veterinary Institute	Norway	WP4-Partner
10	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP2-Partner
11	NOVA	UNIVERSIDADE NOVA DE LISBOA	Portugal	WP9 lead



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Authors and Acknowledgements

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List of Acronyms

EU	European Union
KSA	Knowledge, Skills, and Attitudes
VHV	Village Health Volunteer
WHO	World Health Organization
WP	Work Package

Executive Summary

Work package 9 (WP9) of the PANDASIA project aims to identify and understand other potential One Health interventions and analyse citizen perspectives along their journey, i.e., determinants for their engagement in prevention interventions (initial stage) and how the proposed interventions can affect the social and humanistic dimensions (final stage). The specific WP9 objectives are:

- 01. Systematically identify and rank the most relevant and feasible One Health preventive interventions.
- 02. Assess the effectiveness of the VHV training intervention.
- 03. Pilot a Community Empowerment initiative by translating knowledge and evidence generated in 01 and actively engaging the local community in training workshops and co-developing an integrated, community-based intervention.
- 04. Identify and compare factors that affect the engagement of citizens across social groups in the identified One Health interventions (in 01) and the 3PL intervention through a quantitative survey in the study sites.
- 05. Specify and measure social and humanistic impact metrics to capture the changes related to human well-being besides the outcomes achieved through the 3PL intervention

Village Health Volunteers (VHVs) are critical in Thailand's healthcare system, particularly in rural and underserved areas. They act as a bridge between health authorities and communities, promoting health as a whole, surveillance, and disease prevention. The COVID-19 pandemic highlighted the importance of VHVs, as they were instrumental in disease control, symptom screening, and public health education in their communities. This study evaluates the effectiveness of a co-created Pandemic Prevention and Preparedness Literacy (3PL) training program for VHVs in Chanthaburi, Thailand. The primary aim is to assess changes in zoonotic disease and spillover knowledge, communication skills, and skills to assess the household environment for sites relevant to zoonoses, and attitudes related to zoonotic spillover prevention.

We will conduct a randomized control trial to assess the effectiveness of the training program on primary and secondary outcomes. We will recruit 120 VHVs from different villages in Chanthaburi, Thailand. We will equally randomize them into two comparative groups with 60 VHVs each. Group 1 will receive the training first, followed by the outcome assessment, whereas Group 2 will undergo the outcome assessment before exposure to the intervention. This approach ensures that all participants receive the training while enabling a robust evaluation of its immediate impact. The primary outcome is knowledge measured as a test score. The secondary outcome is communication skills and attitudes. We will measure the outcomes using a structured questionnaire administered by a trained interviewer. Statistical

tests, including t-tests and regression analysis, will assess knowledge, skills, and attitudes (KSA) changes.

This protocol will inform tailored interventions to strengthen VHVs' competencies and guide policies to mitigate zoonotic risks in rural Thailand. The training effectiveness evaluation results will help identify specific areas for improvement in VHV training programs and contribute to developing tailored interventions to enhance health literacy to mitigate zoonotic spillover risks in rural communities across Thailand. This study has the potential to inform national health policies and guide future initiatives for VHVs in similar settings.

Introduction & Background

Village Health Volunteers (VHVs)

Village health volunteers (VHVs) have been an integral part of Thailand's healthcare system since their establishment in the 1970s [1], [2]. Their close collaboration with local authorities and government agencies has played a crucial role in improving health as a whole within rural communities across the country [3], [4]. Recognizing their vital contributions, efforts to enhance the integration and cooperation between local governments and VHVs have intensified. This has led to a significant expansion of the program, with the number of volunteers rising to 1.04 million nationwide, including 15,000 based in Bangkok [2]. Each Village Health Volunteer (VHV) is now tasked with promoting health literacy among 15 households, thereby reinforcing community-based health initiatives [5]. Additionally, VHVs receive a stipend of 1,000 Thai baht (approximately US\$32) per month, which was temporarily increased by 50% (an additional 500 baht) during the COVID-19 pandemic [2], [6].

In these daily efforts to improve health outcomes in rural Thailand, Village Health Volunteers (VHVs) engage in a wide range of activities. These include door-to-door visits, providing households with health education, workshops, and essential resources to enhance their well-being. VHVs also organize group screenings, conduct disease prevention campaigns, and develop community plans for disease control. Additionally, they advise community members on the use of medicines, supplements, herbs, and products to prevent the spread of illnesses [6], [7], [8].

The importance of these efforts was particularly evident during the COVID-19 pandemic when VHVs took on even greater responsibilities in health promotion, disease surveillance, and prevention. Their role became essential in managing the pandemic at the community level, with studies showing that VHVs had a 99% potential in preventing COVID-19 outbreaks in their communities [7]. This period also underscored the critical value of regular training, as VHVs who participated in more than five development sessions per year demonstrated significantly higher potential in managing public health crises compared to those who had fewer or no training sessions [7]. During this period, Thailand's VHVs stayed informed and adaptable by utilizing an app specifically designed for use during the outbreak, as well as engaging in social media group chats with primary care health personnel [2].

The hierarchy of Village Health Volunteers (VHVs) in Thailand is closely integrated with the Ministry of Public Health (MoPH) and local healthcare infrastructure. At the national level, the MoPH oversees the entire VHV system, managing recruitment, training, and service protocols. Regionally and provincially, VHVs collaborate with hospitals at various levels, including district and sub-district health-promoting hospitals. At the district level, VHVs report to district hospitals, which supervise their activities, provide ongoing training, and ensure compliance with health policies. Team leaders (prathan) are responsible for task allocation among VHVs and serve as key intermediaries between VHVs and healthcare professionals [9]. This well-defined structure, supported by digital tools, has strengthened the effectiveness of VHVs in delivering public health services at the community level.

One Health Approach

The One Health approach is a holistic framework that recognizes the interconnected health of humans, animals, and the environment, emphasizing interdisciplinary collaboration to address complex health challenges such as zoonotic diseases and it has been applied in diverse contexts, including antimicrobial resistance, food safety, and zoonotic disease management [10]. Thailand has taken innovative steps in this area, establishing a specific governmental unit for One Health, the Coordinating Unit for One Health (CUOH), within its central government [11]. The Saen Thong One Health Observatory in Nan Province, Thailand, exemplifies a decade-long application of the One Health approach, addressing issues such as rodent-borne diseases, antimicrobial resistance, and pesticide exposure, emphasizing ethical principles, community engagement, and the co-creation of solutions tailored to local contexts, influencing local health policies, developing nature-based solutions, and enhancing community capacity through training programs [12]. Initiatives such as the Thai One Health Network enhance intersectoral coordination, focusing on surveillance, awareness campaigns, and technical training, with persisting challenges, including limited laboratory capacity, human resource constraints, and reliance on external funding [10].

VHVs are integrated into Thailand's One Health approach, acting as community coordinators responsible for raising awareness about the interconnected risks of human, animal, and environmental health [13]. In Thailand, the One Health approach has been prominently applied to tackle avian influenza (AI). VHVs in Thailand play a pivotal role in implementing a community-based surveillance system for AI under the One Health framework. Their efforts are supported by specialized training and regular capacity-building activities, engaging in risk communication, educating villagers about disease prevention, and encouraging early reporting [13]. By fostering collaboration between the Ministry of Public Health, the Ministry of Agriculture, and Surveillance Rapid Response Teams, the system ensures rapid containment of outbreaks and enhances community resilience, making VHVs an essential component of Thailand's public health strategy [13]. Furthermore, in 2016, Thailand piloted an AI surveillance system that integrates stakeholders from the human, animal, and environmental sectors at both the central level and in four provinces. This system monitors influenza A viruses across human, waterfowl, and poultry populations. The AI pilot serves as a strategic framework for developing and enhancing One Health surveillance systems aimed at tackling emerging zoonotic and global pathogens [14].

Objectives

This protocol evaluates the outcomes of a co-created Pandemic Prevention and Preparedness Literacy (3PL) training program for Village Health Volunteers (VHVs) in Chanthaburi, Thailand. It addressed the impact of training programs designed to equip Village Health Volunteers (VHVs) with the knowledge, skills, and attitudes (KSA) to prevent zoonotic spillover risks. More specifically, it assesses changes in zoonotic disease and spillover knowledge, communication skills, skills to assess the household environment for sites relevant to zoonoses, and attitudes about zoonotic spillover prevention. The outcomes are categorized into primary and secondary. The primary outcome of this trial is knowledge, and the secondary

outcomes are skills and attitudes. The evaluation aims to answer the following questions:

1. Does the training intervention significantly enhance VHVs trainee's knowledge and skills in delivering support to the village populations?
2. Does the training intervention significantly influence or change VHVs' attitudes toward providing support to the population?
3. How do VHVs perceive the impact of the training intervention on their performance in providing support to the population?
4. Do VHVs retain the knowledge gained from the training intervention?
5. How do VHVs perceive the impact of the training intervention on their performance in providing support to the population?

Methods

Study Workflow

This protocol is a key phase of a larger project conducted in Chanthaburi, Thailand, which aims to empower Village Health Volunteers (VHVs) with essential knowledge about zoonotic spillover risks and pandemic prevention. The project comprises multiple stages, some of which have been completed, including a qualitative study conducted with community members, Animal Health Volunteers (AHVs), and VHVs. This initial phase was designed to determine the most effective methods for delivering information to these communities.

We employ an experimental design approach to evaluate the effectiveness of VHVs training in Thailand for zoonotic spillover prevention and the knowledge, skills, and attitudes acquired through these training programs. This protocol involves implementing a co-creation training program for Village Health Volunteers (VHVs) in Thailand, focusing on pandemic prevention and preparedness literacy and quantitative data collection through a questionnaire. A random sample of VHVs is drawn from an existing database and participants are randomly assigned into two groups. The first group will receive the training first, followed by the test, while the second group will take the test first and then receive the training. The responses from both groups will be analyzed through statistical tests and linear regression to evaluate the impact of the training on enhancing knowledge, skills, and attitudes related to zoonotic spillover risks and prevention. A graphical representation of this workflow is provided in Figure 1.

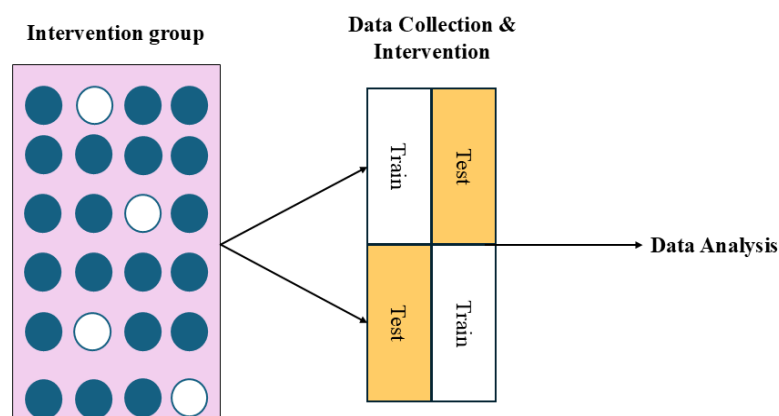


Figure 1. Study workflow.

Study Design

Experimental design in social sciences focuses on understanding the mechanisms implied by observed relationships rather than structural causal explanations, emphasizing the use of methods like randomization, control groups, and statistical inference to isolate specific causal factors and strengthen the validity of conclusions while minimizing spurious effects [15]. We employ experimental design utilizing a randomized controlled trial (RCT). Our protocol adopts a between-subjects design with randomization, where participants are randomly assigned to one of two groups to ensure unbiased comparisons. Group 1 receives the training intervention first, followed by a test to assess the immediate impact of the training. Conversely, Group 2 completes the test first, serving as a baseline measure, and then undergoes the training intervention. This sequence ensures that both groups ultimately receive the training while enabling a comparison of the immediate effects of the intervention on knowledge, skills, and attitudes. The randomized allocation minimizes selection bias, while the structured sequence of training and testing allows for a robust evaluation of the intervention's efficacy. Data from both groups are analyzed to assess the differences in outcomes, isolating the effect of the training intervention in a controlled and systematic manner. This ensures a between-subjects design with randomization reinforcing the experimental study's robustness.

The training intervention comprises the co-created Pandemic Prevention and Preparedness Literacy program (3PL) designed to enhance VHVs' capabilities in zoonotic spillover risk mitigation. Knowledge, skills, and attitudes (KSA) are measured using a standardized questionnaire to evaluate the effectiveness of the training. This robust methodology provides a framework for replication in similar contexts or regions, enhancing the generalizability of the findings. The study will be conducted in collaboration with a multidisciplinary team of researchers, health professionals, and local community members. Ethical approval has been obtained from NOVA IMS in Lisbon and KKU in Thailand. All participants will provide informed consent before participation.

Study Population

The study population will consist of adults (≥ 18 years old) who are actively serving as Village Health Volunteers (VHVs) in Chanthaburi, Thailand. Exclusion criteria will include VHVs who are not willing to respond to the questionnaire due to personal or ethical reasons, as well as those who are unable to speak or read in Thai or English or are completely unable to respond to the questionnaire, either directly or with the assistance of someone.

Study Sample

A randomized sampling approach will be used to assign participants to two groups. The sample will consist of 120 VHVs, randomly selected from an existing database, ensuring equal representation in each group. Randomization will occur at the individual level to minimize selection bias and ensure a fair distribution of participants. This sample size accounts for a 5% non-response rate, ensuring robust representation. Alternates will be pre-identified to address potential problems. This approach allows for a robust and unbiased evaluation of the intervention's effects.

This study involves comparing two groups: one that receives training first (training-test group) and one that receives the test first (test-training group). A sample size calculation was conducted to ensure sufficient statistical power for detecting a 30% change in Knowledge, Skills, and Attitudes (KSA) resulting from the training intervention. The calculation was based on an expected proportion of participants achieving the desired KSA level post-training ($p_1=70\%$) compared to pre-training ($p_2=40\%$). This calculation accounts for the pooled proportion ($p=55\%$) and the minimum detectable difference ($\Delta=30\%$). The formula for determining the required sample size for each group is given by:

$$n = \frac{2(Z_{\alpha/2} + Z_{\beta})^2 \cdot p(1-p)}{(p_1 - p_2)^2}, \text{ where}$$

n = sample size per group

$Z_{\alpha/2}$ = critical value for the desired level of significance (1.96 for 95%)

Z_{β} = power of the test (1.28 for 90% power)

p = baseline proportion (0.55)

p_1 and p_2 = proportions for the pre- and post-training groups, respectively (0.30)

Using a significance level of 0.05 and a power of 90%, the required sample size was determined to be 58 participants per group. To guarantee that all needed participants attend the training, we set a minimum sample size of 60 participants per group, ensuring that the study has the statistical power to detect meaningful changes in KSA outcomes, where:

1. Group 1 (60 participants) will receive the training first, followed by the test.
2. Group 2 (60 participants) will take the test first and then receive the training.

This approach ensures that all participants ultimately receive the training while enabling a robust assessment of its immediate impact. This robust sampling and study design supports the validity of the comparisons between the two groups.

Quantitative Study

Measures

To comprehensively assess the knowledge, skills, and attitudes gained by Village Health Volunteers (VHVs) from the training on zoonotic spillover risk, we will employ a questionnaire as the primary data collection tool. All measures are adapted from validated scales in existing literature, ensuring reliability and validity. Each question within the questionnaire will be specifically tailored to the VHVs' context and the 3PL intervention, to ensure the instrument is relevant and targeted. The factors (knowledge, skills, and attitudes) will be directly aligned with the training content, enabling a precise evaluation of the competencies VHVs have developed. The questionnaire will thus be designed.

Knowledge is the primary metric for assessing the effectiveness of the intervention, measured using a test score. This outcome assesses the extent to which the training intervention enhances VHVs' understanding of zoonotic spillover risks. A standardized test comprising multiple-choice and a binomial scale, where the response is correct or incorrect, and will evaluate the participants' knowledge before and after the training. The test was constructed based on and adapts validated scales [16], [17], [18], [19]. The focus on knowledge as the primary outcome highlights its critical role in empowering VHVs to identify, prevent, and mitigate zoonotic spillover risks effectively. Scores will be normalized to provide a percentage that facilitates comparison between participants and groups.

The secondary outcomes are skills and attitudes, which reflect broader competencies and behavioral readiness resulting from the training. These outcomes are measured using a ten-point interval scale ranging from 1 (strongly disagree) to 10 (strongly agree). These scales allow for nuanced responses and help capture varying degrees of confidence and comprehension among the VHVs [20], [21]. The items measuring skills derive from existing works [7], [8], [22], [23], as well as the items for attitudes [22], [24]. Skills represent VHVs' ability to articulate zoonotic risks, deliver public health messages effectively, and engage with community members, and attitudes represent the perceptions, beliefs, and commitment to applying their knowledge. This includes attitudes toward zoonotic spillover prevention, adherence to One Health principles, and motivation to contribute to community health initiatives. By including these secondary outcomes, the study aims to provide a holistic evaluation of the training's impact, recognizing that effective pandemic preparedness requires not only knowledge but also skills and attitudes.

To ensure linguistic accuracy and cultural relevance, the questionnaire was initially developed in English, then translated into Thai, and subsequently back-translated into English. This process is critical to maintaining the fidelity of the questions across both languages, ensuring that the content remains consistent and comprehensible to all respondents. Before

each questionnaire session, volunteers must give their informed consent, which outlines key details about the project, emphasizes the voluntary nature of participation, and assures respondents of the confidentiality and anonymity of their answers. This ethical consideration is crucial for building trust and encouraging honest responses.

Recruitment

VHVs will be contacted directly via email or mobile phone from an existing database of VHVs in Thailand. Alternatively, recruitment may occur directly within village communities, through enlisting by village heads. All potential participants will be given detailed explanations of the study's purpose, timeline, and their rights as participants. Transportation will be provided if needed to ensure accessibility. Participation is entirely voluntary, and non-participation will not lead to further contact attempts. Both groups will undergo a co-created 3PL training program designed to enhance their understanding of zoonotic spillover risks, One Health principles, and pandemic preparedness. The training will include interactive workshops, scenario-based exercises, and resource materials. To ensure adequate sample sizes and in case any participant decides to withdraw consent, alternate VHVs meeting the study's criteria will be pre-identified from the database. Then these alternates will be invited to join the study in their place, allowing us to meet our target while maintaining representation and inclusivity across the sample.

Data Collection

Both primary and secondary outcomes are assessed through a structured questionnaire administered once in two different periods to two different groups. The VHVs in Group 1 will complete the training first, followed by the test, to evaluate the immediate impact of the training intervention on knowledge, skills, and attitudes (KSA). Group 2 will take the test first, serving as a baseline, and then receive the training. This design ensures that both groups ultimately receive the training while enabling a comparison of the immediate effects of the training on KSA outcomes. By alternating the sequence of training and testing, the study aims to assess the effectiveness of the intervention in a systematic and unbiased manner.

The questionnaire will be administered using a tablet provided to the interviewers, ensuring accessibility for all Village Health Volunteers [25]. This approach aims to bridge the digital divide and mitigate any barriers to participation caused by a lack of electronic devices. By facilitating access to the questionnaire in this manner, we aim to reduce non-sampling bias in our study and promote more inclusive participation. Our project partners from Thailand, who are fluent in Thai, will handle the translation of the research instruments, including the questionnaire, and will be present in the fieldwork. This ensures the accuracy and cultural appropriateness of the materials used in the study, facilitates effective communication, and ensures the validity of the data collected from Thai-speaking participants.

Upon agreeing to participate, VHVs will be invited to their local health district center to complete the survey, with transportation provided if needed. At the center, each VHV will complete a single data collection session lasting approximately 15 to 30 minutes. The data collection period is anticipated to span three months. The project partners from Thailand will

maintain regular contact with VHVs as needed to address questions, provide support, and ensure a smooth process. When a VHV chooses to withdraw the informed consent and no longer participate, we will terminate the data collection session.

Data analysis

For the quantitative analysis in this study, experimental design techniques will be utilized to assess the significance of differences observed between the two groups of VHVs. This comparison aims to determine whether meaningful differences indicate the training program's effectiveness in enhancing VHVs' knowledge, skills, and attitudes (KSA) related to managing zoonotic spillover risks. Ultimately, the analysis will evaluate whether the training intervention better equips VHVs to take appropriate actions in protecting the communities they serve.

- Primary outcomes - Knowledge:

The knowledge items will be scored as binary (1 for correct answers, 0 for incorrect answers). These scores will be summed and normalized by dividing by the total number of knowledge questions, yielding a score as a percentage. The desired improvement in knowledge is represented by a 30% increase in test scores from pre-training to post-training. For example, if the baseline knowledge score averages 40%, the desired post-training score would be 70% or above. This benchmark indicates that VHVs have achieved a robust understanding of: Zoonotic diseases and their transmission pathways, one Health principles and their practical applications, and effective zoonotic spillover prevention strategies.

- Secondary outcomes – Skills & Attitudes

For skills and attitudes, which are rated on a 10-point interval scale (1 = strongly disagree, 10 = strongly agree), the average score for each participant will be computed, this allows us to quantify each participant's overall skills and attitudes [16]. For skills, the desired improvement entails a 30% improvement in VHVs' ability to effectively communicate zoonotic spillover risks, conduct assessments of households and environments for zoonotic risks, and confidently apply intervention protocols in community settings. This improvement is measured using a 10-point scale, where the average score increases from a baseline (e.g., 5) to at least 7 or above post-training.

The desired level of attitudes reflects a 30% positive shift in participants' perceptions and motivation, as measured by a greater commitment to zoonotic spillover prevention, enhanced appreciation for One Health principles, and increased motivation to fulfill their role as VHVs. Similarly, this improvement is benchmarked on a 10-point scale, where scores rise from a baseline (e.g., 5) to 7 or above.

By defining the "desired KSA level" as a 30% improvement across all components, this study sets measurable and achievable goals for the intervention, ensuring alignment with the objectives of evaluating the training's effectiveness. Outliers will be identified using standardized residuals or interquartile range methods. Any identified outliers will be reviewed

contextually before exclusion. Descriptive statistics will be provided to present sociodemographic characteristics using means, standard deviations, medians, and frequencies.

To evaluate the impact of the training, inferential statistical tests will be applied. For all statistical tests, a significance level of 5% ($p < 0.05$) will be used to determine statistical significance. For the knowledge-related questions, a Chi-squared test will be used to compare the number of correct and incorrect responses from pre and post-tests, allowing us to determine whether the training has had a significant impact on participants' knowledge. Given that knowledge items are binary, this test is appropriate for assessing differences between categorical outcomes across groups. For skills and attitudes, logistic regression will be used. Also, independent t-tests will be conducted to compare the mean scores between VHV participants who participated in the training and those who did not. These tests will help assess whether the training program significantly impacted the skills and attitudes of the participants. In addition to these analyses, correlation and regression analyses will be conducted to explore the relationships between knowledge, skills, and attitudes [25], [26]. The Pearson correlation coefficient will be computed to check for linear relationships between knowledge, skills, and attitudes, allowing us to see if changes in one area correspond to changes in another.

Regression analysis will be used to understand the impact of factors such as gender, age, district, literacy level, and participation in the training program on the KSA scores. These factors will serve as associated factors in the regression models, which will help assess whether variations in KSA scores are statistically significant concerning these associated factors. Given the relatively small sample size, multiple linear regression is a feasible approach if there are enough data points per determinant to avoid overfitting and ensure stable estimates [27]. The expected effect size should be moderate, as we anticipate a noticeable but not overwhelming impact of the training on the KSA components. The regression models will be evaluated to determine the strength and direction of the relationships between associated factors and outcomes. All t-tests will be conducted only if normality and variance homogeneity assumptions are met. Non-parametric tests will be considered otherwise.

Ethical Considerations

This study received approval from NOVA IMS's ethics committee in Lisbon and follows KKU's guidelines in Thailand. To enhance transparency, we pre-registered the study on the platform AsPredicted. Participants' confidentiality and anonymity will be safeguarded throughout the study. Data will be aggregated during analysis to ensure no individual responses are identifiable. Measures will be taken to address potential ethical concerns in sampling, ensuring respect for participant privacy.

Before obtaining consent, participants will receive comprehensive information about the study in written format. Informed consent will be obtained from all participants, either in written or verbal form. All data collected will be completely anonymous, ensuring participant confidentiality. The importance of anonymity will be emphasized to encourage honest and candid responses [26]. No individual responses will be disclosed during the quantitative study.

Only aggregated data will be shared after analysis, ensuring that no individual participant can be identified. This includes demographic information, such as age, gender, and other relevant characteristics, which will also be presented in an aggregated form to safeguard participant anonymity.

Next Steps & Dissemination Plan

The questionnaire for assessing the primary (knowledge) and secondary outcomes (skills and attitudes) is still under development. Thus, the version presented in the appendices is not final and is yet to be improved based on the literature and experts' contributions. Data collection will begin in the dry season of 2025, followed immediately by the data analysis phase. Subsequently, the quantitative data will be analyzed, with results prepared for submission in top-tier peer-reviewed journals.

Appendices

Appendix 1: Questionnaire for assessing knowledge, skills, and attitudes

Question			Scale	Correct answer	Reference
Knowledge					
1	When animals are sick in your flock, you can get the same sickness.		Correct / Incorrect	Correct	[16]
2	Many animal diseases are transmitted from humans to animals			Incorrect	
3	Animal disease can only be transmitted through direct contact.			Incorrect	
4	Eating uncooked meat is always safe and free from disease risks.			Incorrect	
5	Drinking of raw milk cannot transmit diseases from animals to you			Correct	
6	Close contact with a sick or dead animal is completely safe and free from disease risks.			Incorrect	
7	You can get infections from an environment contaminated by secretions of sick animals			Correct	
8	Animal bites are harmless and free from any risk of disease transmission.			Correct	
9	Animal bites cannot transmit diseases to you			Incorrect	
10	Assisting animals during parturition with bare hands is safe and hygienic.			Incorrect	
11	Assisting newborns right after delivery exposes you to diseases			Correct	
12	Contact with aborted materials poses no risk of disease			Incorrect	
13	Collecting aborted fetuses and placenta with bare hand exposes you to diseases			Correct	
14	Tuberculosis	a. can spread to humans and cattle by eating raw/uncooked pork and fruits/vegetables infected with eggs of parasites in human excreta	Choose the correct answer	14e	[17]
15	Plague	b. can spread to humans by direct contact with dogs and cats		15c	
16	Rabies	c. can spread to humans by the bite of a flea		16f	
17	Swine Flue	d. can spread to humans by consuming food infected with eggs of parasites from dog excreta		17g	
18	Taeniasis	e. can spread to humans through aerosols, raw milk, raw/uncooked meat, polluted environment		18a	
19	Hydatidosis	f. can be spread to humans by the bite of a rabid dog		19d	
20	Ringworm	g. can spread to humans by eating pork and/or chicken		20b	

21	Human Avian Influenza is an infectious disease	Correct / Incorrect	Correct	[18]
22	Humans can only be infected with Avian Influenza virus from direct contact with other humans		Incorrect	
23	Humans may not fully recover from infection with Avian Influenza virus after treatment		Correct	
24	Human infection with the Avian Influenza virus is unrelated to hygiene in wet poultry markets.		Incorrect	
25	Avian Influenza is the same as fowl plague (Newcastle disease)		Incorrect	
26	Dengue is transmitted by mosquito bite	Correct / Incorrect	Correct	[19]
27	Dengue virus can be transmitted from infected pregnant mother to the baby		Correct	
28	Dengue infection significantly reduces the risk of miscarriage		Incorrect	
29	Can all mosquitoes transmit dengue?		Incorrect	
30	Aedes mosquitoes prefer to breed in saltwater environments like oceans and seas.		Incorrect	
31	Aedes mosquito can breed both indoor and outdoor		Correct	
32	Clean water as little as 5mL is enough for mosquitoes to breed		Correct	
33	Mosquitoes require a minimum of 30 days for complete maturation from eggs to adults		Incorrect	
34	Mosquitoes like to bite early in the morning and late evening		Correct	
35	Dengue fever is a mild illness and never results in death		Correct	
36	People who have had dengue became immune to the disease		Incorrect	
37	Assisting the animal in delivery with bare hands can expose you to disease risks.	Correct / Incorrect	Correct	[16]
38	Collecting the aborted fetuses and placenta with bare hands is completely safe and poses no disease risks		Incorrect	
39	You are at risk of acquiring diseases from abortion-causing agents		Correct	
40	Many of the agents that cause abortion in animals are harmless.		Incorrect	
41	Animal health care providers can handle abortion outbreaks very well		Correct	
42	The spread of animal abortion-causing agents to humans is unavoidable		Incorrect	

Question		Scale	Reference
Skills			
1	I can accurately identify geographical regions experiencing a specific zoonotic disease outbreak	1 - Strongly Disagree, 10 - Strongly Agree	[22]
2	I can utilize epidemiological tools and risk assessment models to evaluate the likelihood of zoonotic spillover from wildlife to livestock in specific habitats		
3	I am capable of designing an intervention or change to avoid a zoonotic spillover		

4	I am capable to act accordingly during an intervention to avoid a zoonotic spillover		
5	I am capable to use a set of measures to evaluate the impact of the intervention		
6	I can identify environmental and socio-economic factors that increase the vulnerability of communities to zoonotic spillover events, particularly in high-risk areas like urban-wildlife interfaces		
7	I can gather and analyze relevant data from scientific literature, surveillance reports, and public health databases to understand zoonotic spillover incidents		[7]
8	I am capable to read and share information about zoonotic spillover		[7]
9	I am capable of writing messages for others to understand		
10	I can communicate zoonotic spillover risks through home visits and community meetings to raise awareness among local populations.		[8]
11	Understanding & using the protocols is easy when managing a suspected/confirmed patient		
12	Understanding my role in the protocols is easy when managing a suspected/confirmed patient		
13	I am aware of resources or online material to use for finding out more about VHF		
14	I am aware of the RMH infection prevention resources		

Question		Scale	Reference
Attitudes			
1	You frequently wash your hands with soap to reduce cross-contamination during an outbreak.	1 - Strongly Disagree, 10 - Strongly Agree	[24]
2	You often use a face mask to reduce the spread of disease during an outbreak.		
3	You usually use mass transport during an outbreak.		
4	You often engage in outdoor activities with friends during an outbreak.		
5	You usually maintain social distancing to reduce the spread of disease during an outbreak.		
6	You adhere to health policies to reduce the spread of disease during an outbreak.		
7	You usually eat street snacks during an outbreak.		
8	You often clean your home and office during an outbreak.		
9	You regularly deal with sick people or health workers during an outbreak.		
10	You usually share food or water with others during an outbreak.		
11	I am interested in Village Health Volunteering (VHV)		[22]
12	I understand the role VHV plays in reducing zoonotic spillover risk		
13	VHV plays an important role in strengthening systems, such as healthcare		
14	I value VHV training as part of my citizenship development		

15	I want to participate in VHV initiatives as an integrated member of the community		
16	The VHV training can help make a change in my national healthcare system		
17	Using my VHV training in the real world will make improvements		
18	I understand the rationale for VHV in the real world		
19	I intend to use the skills I learn in my VHV training to make a difference in my community		

Question		Scale	Reference
Motivation			
1	I feel motivated to contribute to the One Health intervention	1 - Strongly Disagree, 10 - Strongly Agree	[28]
Effectiveness			
1	I understand the complex and difficult parts of zoonotic spillover	1 - Strongly Disagree, 10 - Strongly Agree	[28]
2	I understand the complex and difficult parts of the One Health approach		

References

- [1] S. D. Kowitt, D. Emmerling, E. B. Fisher, and C. Tanasugarn, "Community Health Workers as Agents of Health Promotion: Analyzing Thailand's Village Health Volunteer Program," *J Community Health*, vol. 40, no. 4, pp. 780–788, Aug. 2015, doi: 10.1007/s10900-015-9999-y.
- [2] M. Narkvichien, "Thailand's 1 million village health volunteers - 'unsung heroes' - are helping guard communities nationwide from COVID-19," World Health Organization. Accessed: Nov. 21, 2024. [Online]. Available: <https://www.who.int/thailand/news/feature-stories/detail/thailands-1-million-village-health-volunteers-unsung-heroes-are-helping-guard-communities-nationwide-from-covid-19>
- [3] M.-R. Miri, A. A. Ramazani, M. Moodi, and K. Mirkarimi, "The effects of suburban villages' health volunteer plan on women's health knowledge and attitude," *Journal of Education and Health Promotion*, vol. 1, no. 1, p. 12, 2012, doi: 10.4103/2277-9531.98570.
- [4] P. Jiaviriyaboonya, "Anthropological study of village health volunteers' (VHVs') socio-political network in minimizing risk and managing the crisis during COVID-19," *Heliyon*, vol. 8, no. 1, p. e08654, Jan. 2022, doi: 10.1016/j.heliyon.2021.e08654.
- [5] C. Jewjinda and N. Chalermnirundorn, "The Development of Village Health Volunteers (VHVs) Model with a Participatory Process," 2018.
- [6] W. Chankong and P. Polprasert, "Potential of Village Health Volunteers (VHVs) in Preventing COVID-19 in Rural Thailand," *Journal of Pharmaceutical Negative Results*, vol. 13, no. 4, pp. 1506–1513, Dec. 2022.
- [7] Wongkongdech *et al.*, "Lessons learned and the potential of thai vhvs on the effectiveness of covid-19 control: a cross-sectional study," *Journal of Management Information and Decision Sciences*, vol. 25, no. 1, Dec. 2021, Accessed: Nov. 21, 2024. [Online]. Available: <https://www.abacademies.org/abstract/lessons-learned-and-the-potential-of-thai-vhvs-on-the-effectiveness-of-covid19-control-a-crossectional-study-13862.html>
- [8] S. Visanuyothin, C. Chatanuluk, S. Saengsuwan, A. Rojanavarapong, and P. Pornchanya, "Health Literacy of Village Health Volunteer in Municipality, Nakhonratchasima, Thailand," *Journal of Public Health and Development*, vol. 13, no. 1, Art. no. 1, Jul. 2015, doi: 10.14456/jphd.2015.6.
- [9] A. Cohen and P. T. Cohen, "Paradigm Conflict: Village Health Volunteers and Public Health in Thailand," *The Asia Pacific Journal of Anthropology*, vol. 25, no. 3, pp. 251–265, May 2024, doi: 10.1080/14442213.2024.2357104.
- [10] B. D. Rai, G. A. Tessema, L. Fritschi, and G. Pereira, "The application of the One Health approach in the management of five major zoonotic diseases using the World Bank domains: A scoping review," *One Health*, vol. 18, p. 100695, Jun. 2024, doi: 10.1016/j.onehlt.2024.100695.
- [11] R. Tangwangvivat *et al.*, "Collective activities of the Thai Coordinating Unit for One Health (CUOH): Past activities and future directions," *One Health*, vol. 18, p. 100728, Jun. 2024, doi: 10.1016/j.onehlt.2024.100728.
- [12] K. Chaisiri *et al.*, "A Social-Ecological and One Health Observatory: Ten Years of Collaborative Studies in Saen Thong (Nan, Thailand)," *One Health Cases*, vol. 2023, p. ohcs20230008, Feb. 2023, doi: 10.1079/onehealthcases.2023.0008.
- [13] W. H. O. R. O. for S.-E. Asia, "Role of village health volunteers in avian influenza surveillance in Thailand," Art. no. SEA-CD-159, 2007, Accessed: Nov. 27, 2024. [Online]. Available: <https://iris.who.int/handle/10665/205876>

- [14] G. K. Innes *et al.*, "Enhancing global health security in Thailand: Strengths and challenges of initiating a One Health approach to avian influenza surveillance," *One Health*, vol. 14, p. 100397, Jun. 2022, doi: 10.1016/j.onehlt.2022.100397.
- [15] G. Julnes, "Experiments, Overview," in *Encyclopedia of Social Measurement*, K. Kempf-Leonard, Ed., New York: Elsevier, 2005, pp. 885–899. doi: 10.1016/B0-12-369398-5/00001-3.
- [16] G. Alemayehu, G. Mamo, H. Desta, B. Alemu, and B. Wieland, "Knowledge, attitude, and practices to zoonotic disease risks from livestock birth products among smallholder communities in Ethiopia," *One Health*, vol. 12, p. 100223, Jun. 2021, doi: 10.1016/j.onehlt.2021.100223.
- [17] B. B. Singh, R. Kaur, G. S. Gill, J. P. S. Gill, R. K. Soni, and R. S. Aulakh, "Knowledge, attitude and practices relating to zoonotic diseases among livestock farmers in Punjab, India," *Acta Tropica*, vol. 189, pp. 15–21, Jan. 2019, doi: 10.1016/j.actatropica.2018.09.021.
- [18] N. Xiang *et al.*, "Knowledge, attitudes and practices (KAP) relating to avian influenza in urban and rural areas of China," *BMC Infectious Diseases*, vol. 10, no. 1, p. 34, Feb. 2010, doi: 10.1186/1471-2334-10-34.
- [19] S. Selvarajoo *et al.*, "Knowledge, attitude and practice on dengue prevention and dengue seroprevalence in a dengue hotspot in Malaysia: A cross-sectional study," *Sci Rep*, vol. 10, no. 1, p. 9534, Jun. 2020, doi: 10.1038/s41598-020-66212-5.
- [20] V. Morrison *et al.*, "A Multinational Longitudinal Study Incorporating Intensive Methods to Examine Caregiver Experiences in the Context of Chronic Health Conditions: Protocol of the ENTWINE-iCohort," *International Journal of Environmental Research and Public Health*, vol. 19, no. 2, Art. no. 2, Jan. 2022, doi: 10.3390/ijerph19020821.
- [21] Z. Y. Tang, K. S. Ng, Y. L. E. Koh, and M. T. Yeung, "Study protocol for the validation of a new pictorial functional scale in patients with knee osteoarthritis: the functional activity scoring tool (FAST)," *BMJ Open*, vol. 14, no. 1, p. e076947, Jan. 2024, doi: 10.1136/bmjopen-2023-076947.
- [22] A. Brown, A. Nidumolu, M. McConnell, K. Hecker, and L. Grierson, "Development and psychometric evaluation of an instrument to measure knowledge, skills, and attitudes towards quality improvement in health professions education: The Beliefs, Attitudes, Skills, and Confidence in Quality Improvement (BASiC-QI) Scale," *Perspect Med Educ*, vol. 8, no. 3, pp. 167–176, Jun. 2019, doi: 10.1007/s40037-019-0511-8.
- [23] J. J. Fryk *et al.*, "Knowledge, attitudes and practices of healthcare workers within an Australian tertiary hospital to managing high-consequence infectious diseases," *Infection, Disease & Health*, vol. 26, no. 2, pp. 95–103, May 2021, doi: 10.1016/j.idh.2020.10.002.
- [24] S. M. T. Williams, R. Ansumana, J. Johnny, I. A. Bakarr, and A. Osborne, "Knowledge, Attitudes and Practices Assessment on Bat-Borne Zoonotic Diseases Among the People of Moyamba District, Sierra Leone," *RMHP*, vol. 16, pp. 1331–1342, Jul. 2023, doi: 10.2147/RMHP.S413802.
- [25] M. J. Batalha *et al.*, "Health literacy – study protocol for LiSa cohort study," *BMC Public Health*, vol. 24, no. 1, p. 1737, Jun. 2024, doi: 10.1186/s12889-024-19148-8.
- [26] A. Rehman, T. Saba, K. Haseeb, R. Singh, and G. Jeon, "Smart health analysis system using regression analysis with iterative hashing for IoT communication networks," *Computers and Electrical Engineering*, vol. 104, p. 108456, Dec. 2022, doi: 10.1016/j.compeleceng.2022.108456.

- [27] A. Joinson, "Social desirability, anonymity, and internet-based questionnaires," *Behavior Research Methods, Instruments, & Computers*, vol. 31, no. 3, pp. 433–438, Sep. 1999, doi: 10.3758/BF03200723.
- [28] J.-M. Hu, F.-C. Liu, C.-M. Chu, and Y.-T. Chang, "Health Care Trainees' and Professionals' Perceptions of ChatGPT in Improving Medical Knowledge Training: Rapid Survey Study," *Journal of Medical Internet Research*, vol. 25, no. 1, p. e49385, Oct. 2023, doi: 10.2196/49385.



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